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GRAD 50500

11 September 2026

Week 3 Assignment 1

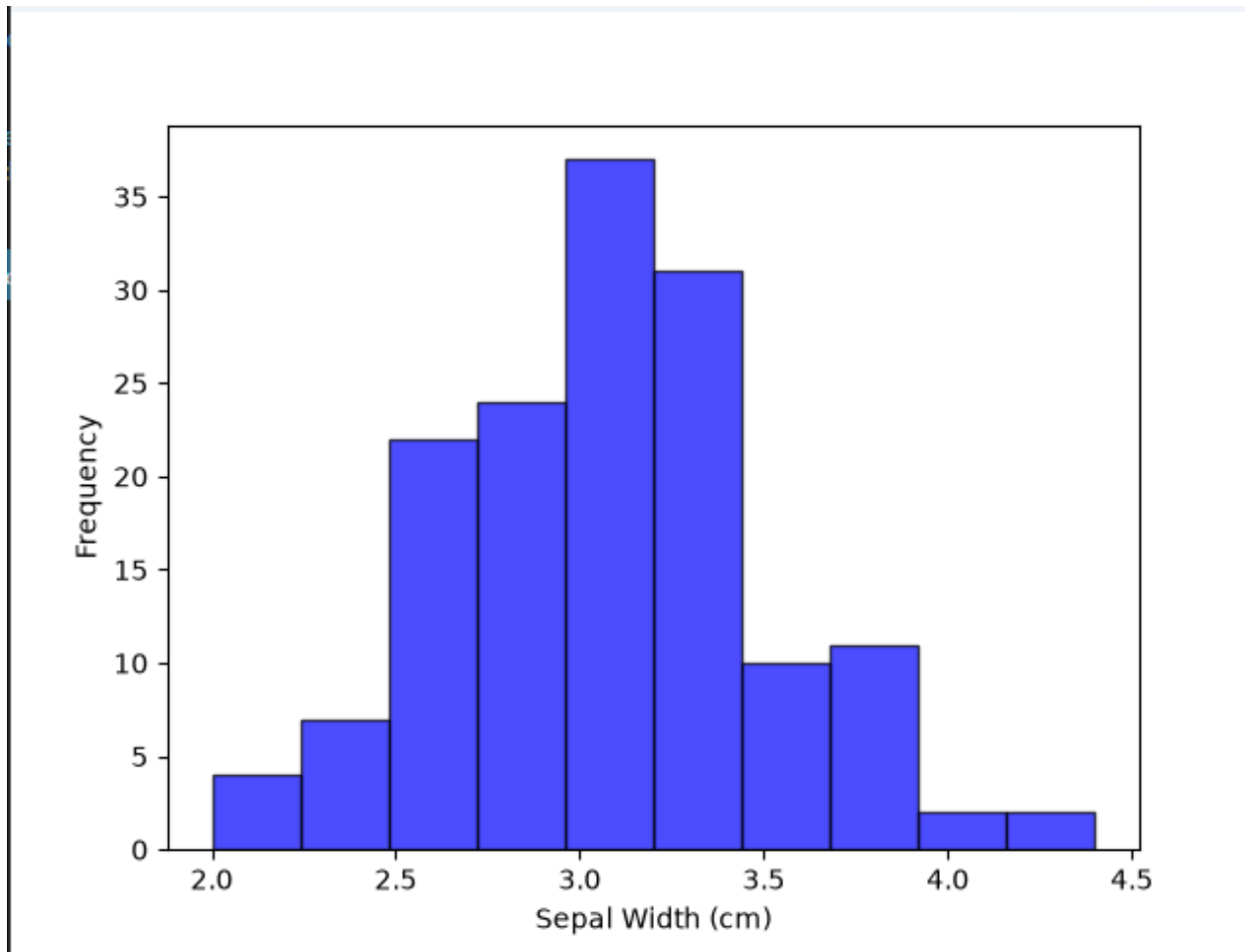
Loading the data:

```
1 from sklearn import datasets
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 import numpy as np
5
6 iris = datasets.load_iris()
7 data = {"weight": [4.37, 5.58, 5.18, 6.11, 4.58, 4.61, 5.17, 4.53, 5.31, 5.14, 4.81, 4.17, 4.41, 3.59, 5.87, 3.83, 6.03, 4.89, 4.32, 4.69, 6.31, 5.12, 5.54, 5.58, 5.37, 5.29, 4.92, 6.15, 5.88, 5.26], "group": ["t1" * 10 + "t2" * 10 + "t3" * 10]}
8 PlantGrowth = pd.DataFrame(data)
```

1.

a.

```
iris_df = pd.DataFrame(iris.data, columns=iris.feature_names)
plt.hist(iris_df["sepal width (cm)"], bins=10, color='blue', alpha=0.7, edgecolor='black')
plt.xlabel("Sepal Width (cm)")
plt.ylabel("Frequency")
plt.show()
```



b. Based on the histogram from 1a. I expect that the mean will be bigger than the median because the tail is more pronounced on the right. If the tail was on the left instead the median would be greater, and if the tails were even the median and mean would be about the same.

```
mean = np.mean(iris_df["sepal width (cm)"])
median = np.median(iris_df["sepal width (cm)"])
print(f"Mean: {mean}, Median: {median}")
```

c.

```
Mean: 3.0573333333333337, Median: 3.0
```

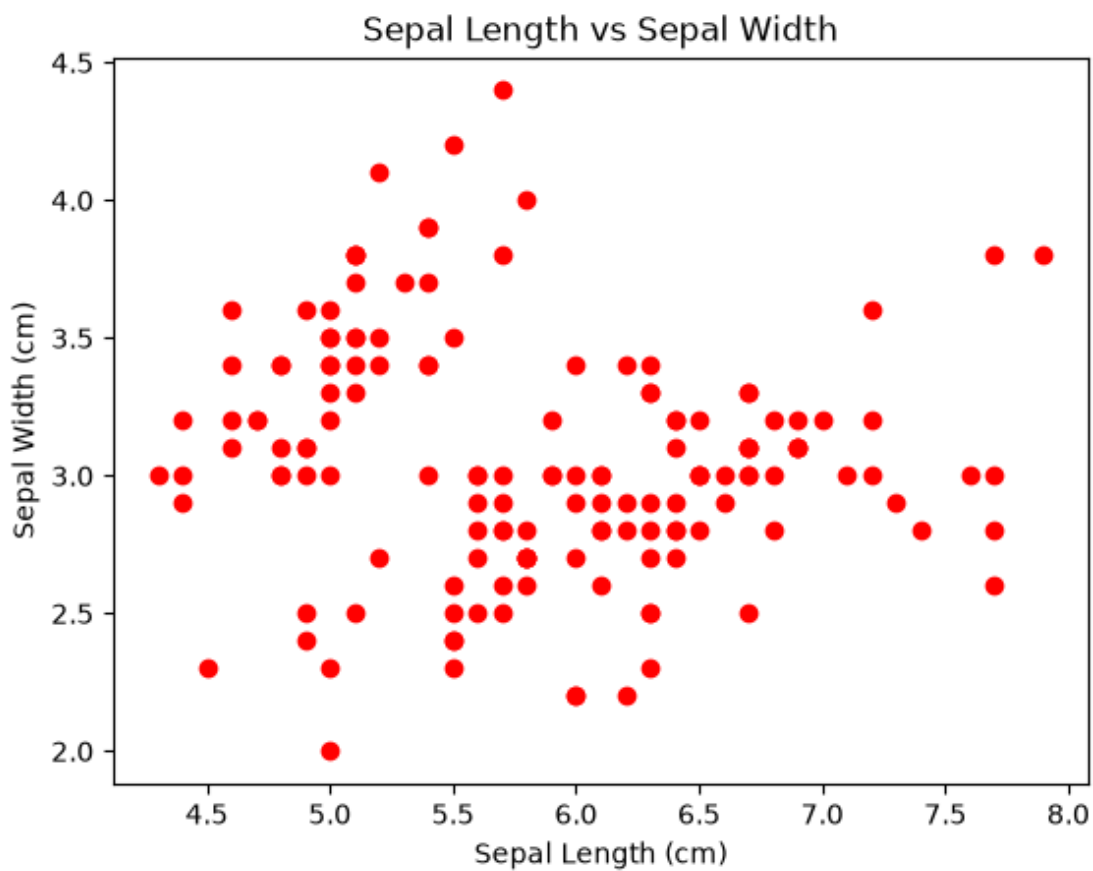
d. 3.3cm because 73rd percentile means that 73% of the flowers were below this number in width, meaning that the top 27% are at or above this number.

```
top27width = np.percentile(iris_df["sepal width (cm)"], [73])  
print(f"73th Percentile: {top27width}")
```

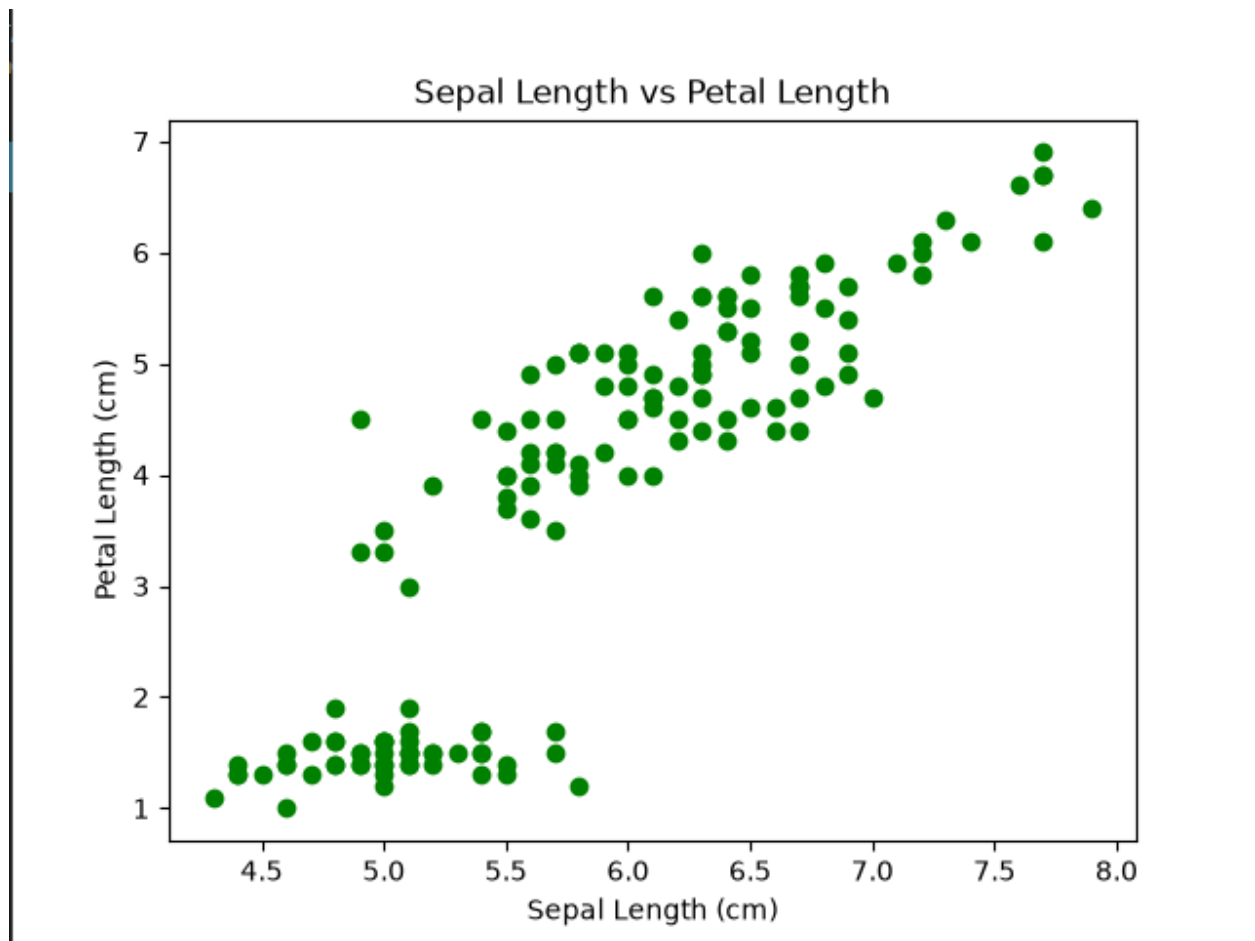
```
73th Percentile: [3.3]
```

e.

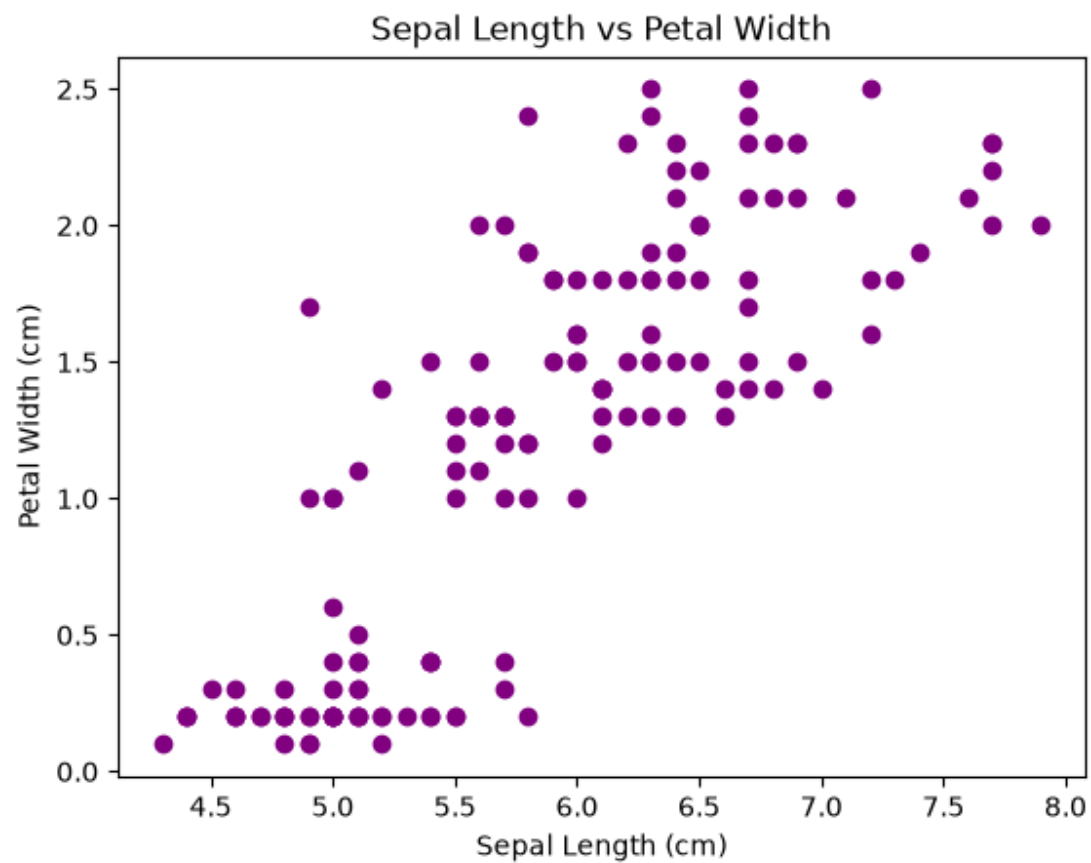
```
plt.scatter(iris_df["sepal length (cm)"], iris_df["sepal width (cm)"], color='red')  
plt.xlabel("Sepal Length (cm)")  
plt.ylabel("Sepal Width (cm)")  
plt.title("Sepal Length vs Sepal Width")  
plt.show()
```



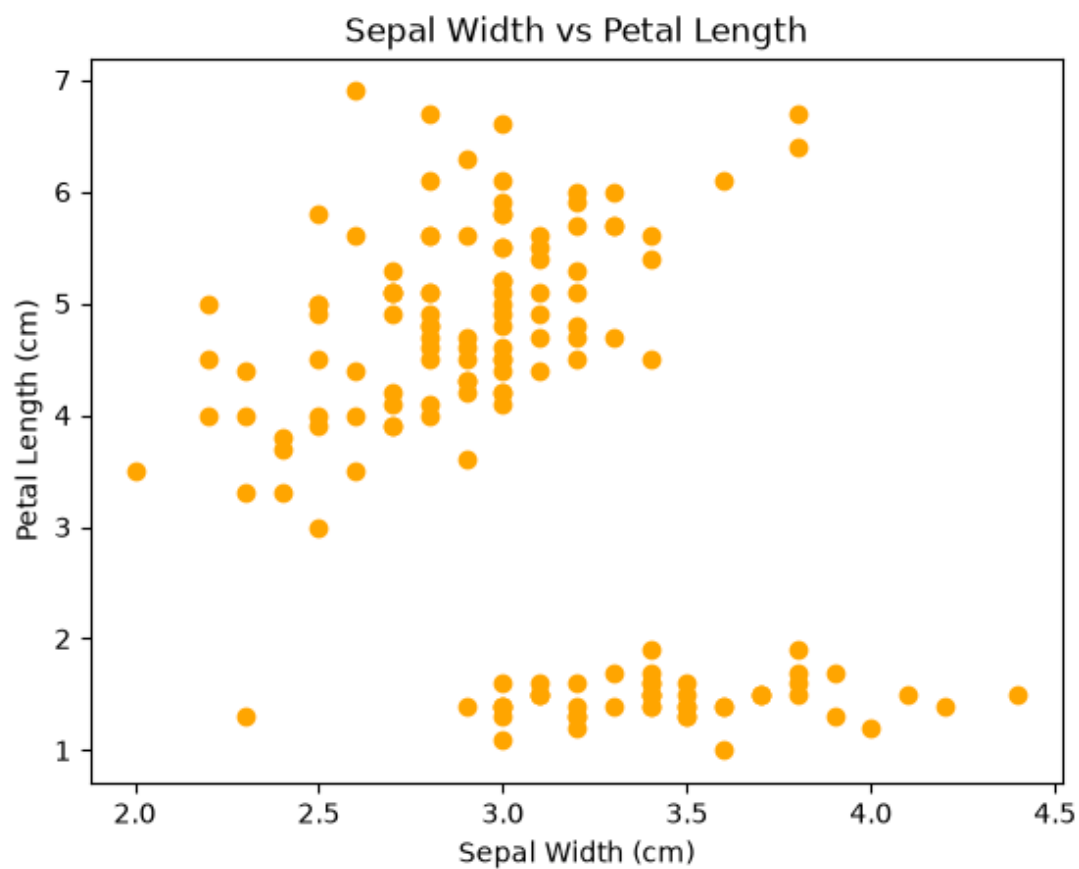
```
plt.scatter(iris_df["sepal length (cm)"], iris_df["petal length (cm)"], color='green')
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.title("Sepal Length vs Petal Length")
plt.show()
```



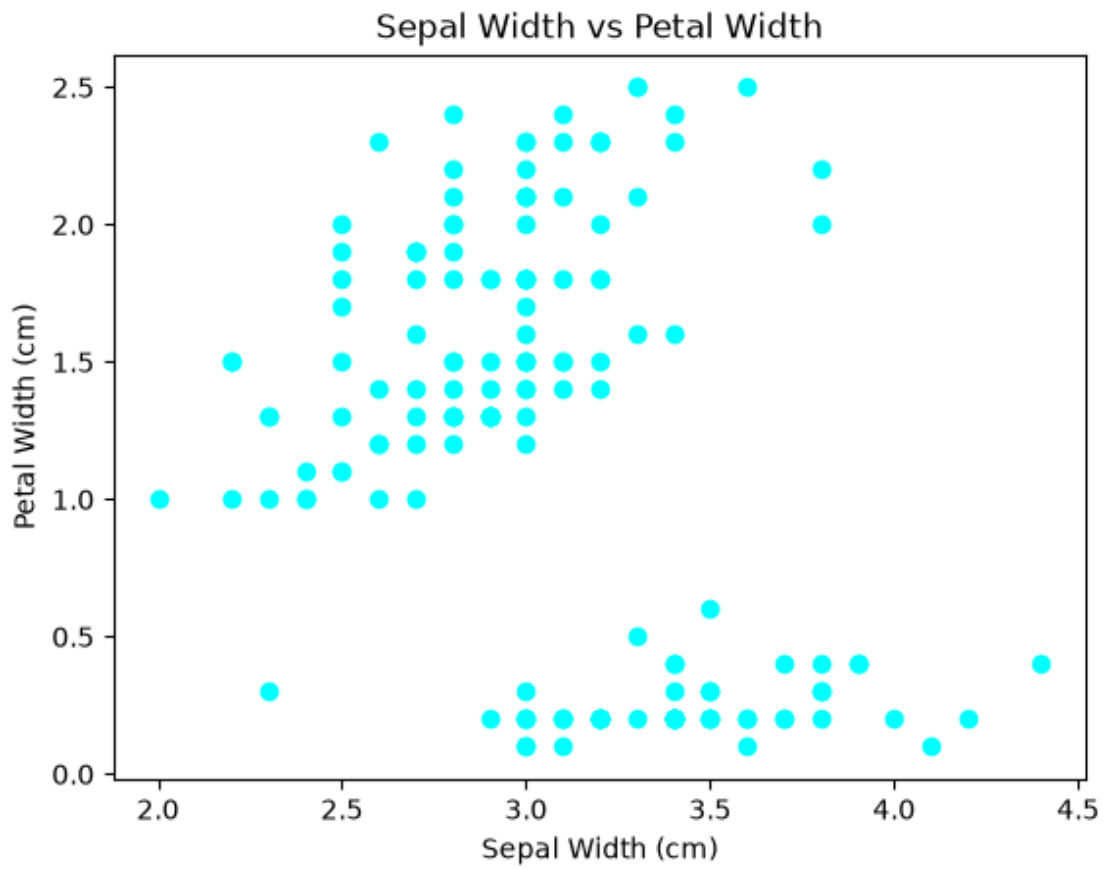
```
plt.scatter(iris_df["sepal length (cm)"], iris_df["petal width (cm)"], color='purple')
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Width (cm)")
plt.title("Sepal Length vs Petal Width")
plt.show()
```



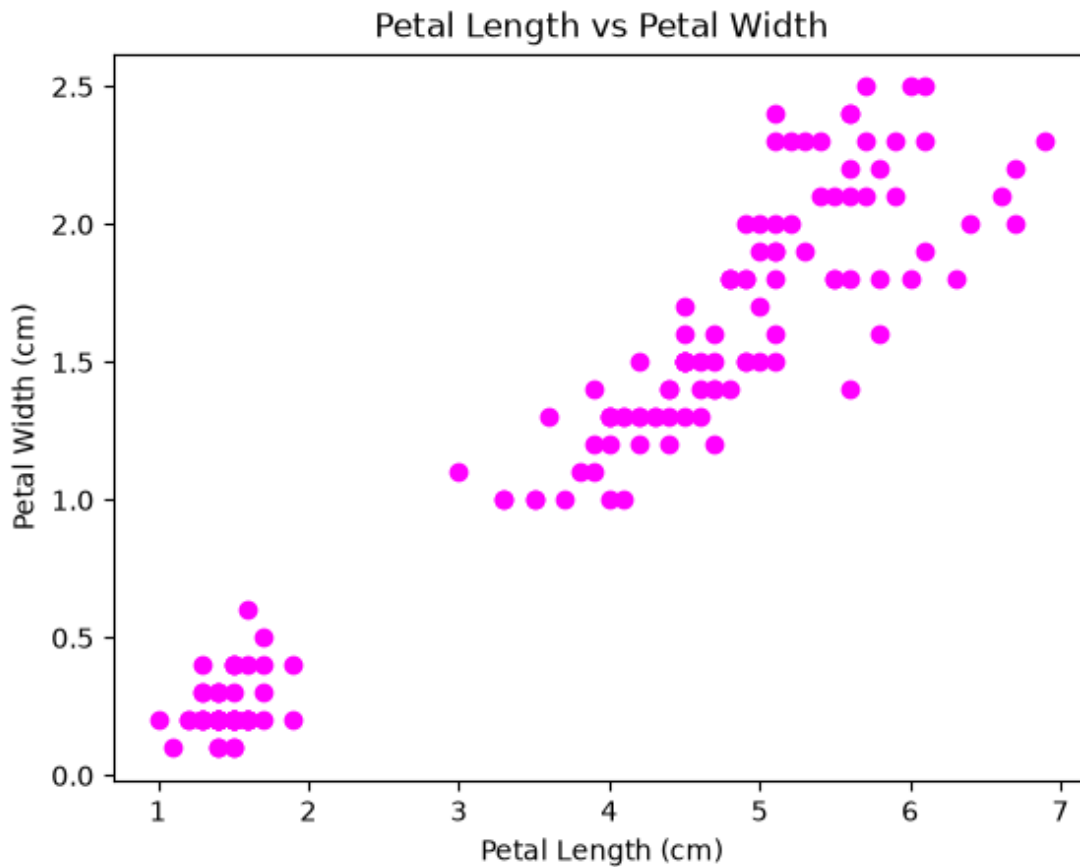
```
plt.scatter(iris_df["sepal width (cm)"], iris_df["petal length (cm)"], color='orange')
plt.xlabel("Sepal Width (cm)")
plt.ylabel("Petal Length (cm)")
plt.title("Sepal Width vs Petal Length")
plt.show()
```



```
plt.scatter(iris_df["sepal width (cm)"], iris_df["petal width (cm)"], color='cyan')
plt.xlabel("Sepal Width (cm)")
plt.ylabel("Petal Width (cm)")
plt.title("Sepal Width vs Petal Width")
plt.show()
```



```
plt.scatter(iris_df["petal length (cm)"], iris_df["petal width (cm)"], color='magenta')
plt.xlabel("Petal Length (cm)")
plt.ylabel("Petal Width (cm)")
plt.title("Petal Length vs Petal Width")
plt.show()
```

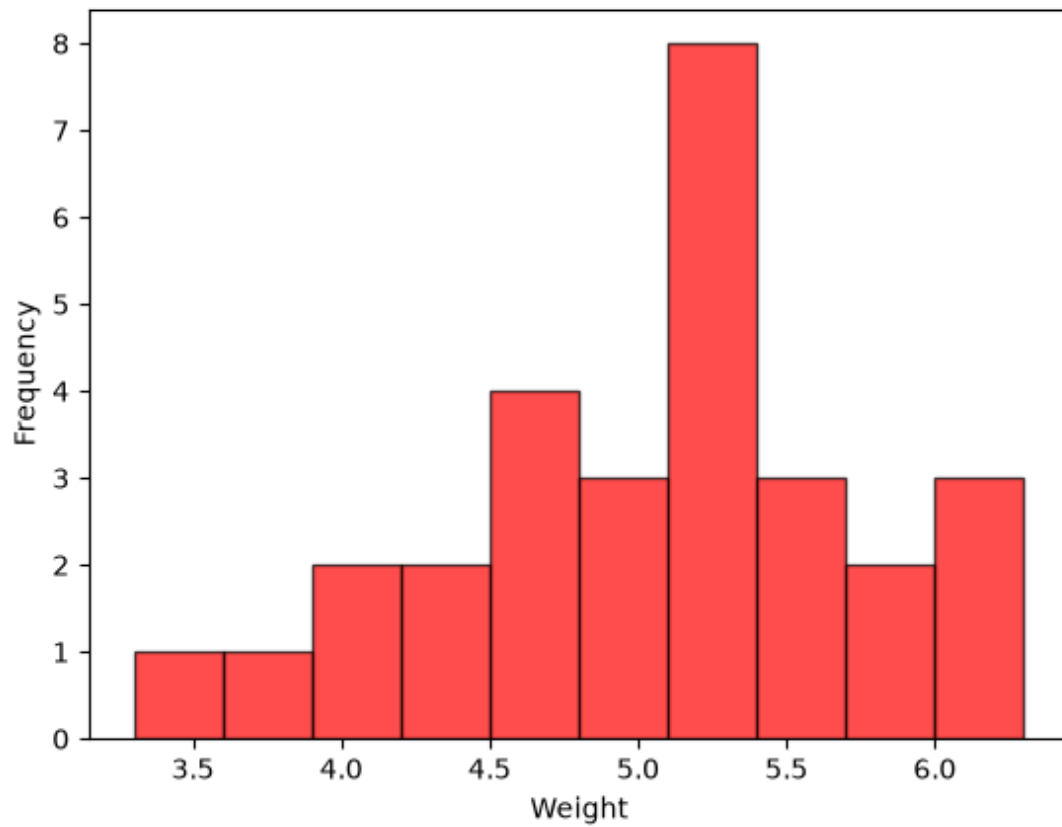


f. Petal Length vs Petal Width has the strongest relationship as the points are the closest together and form a clear upward trend that as petal length goes up so does petal width. Sepal length vs Sepal width seems to have the weakest association as the points form a messy cloud with no clear upward or downward trend.

2.

a.

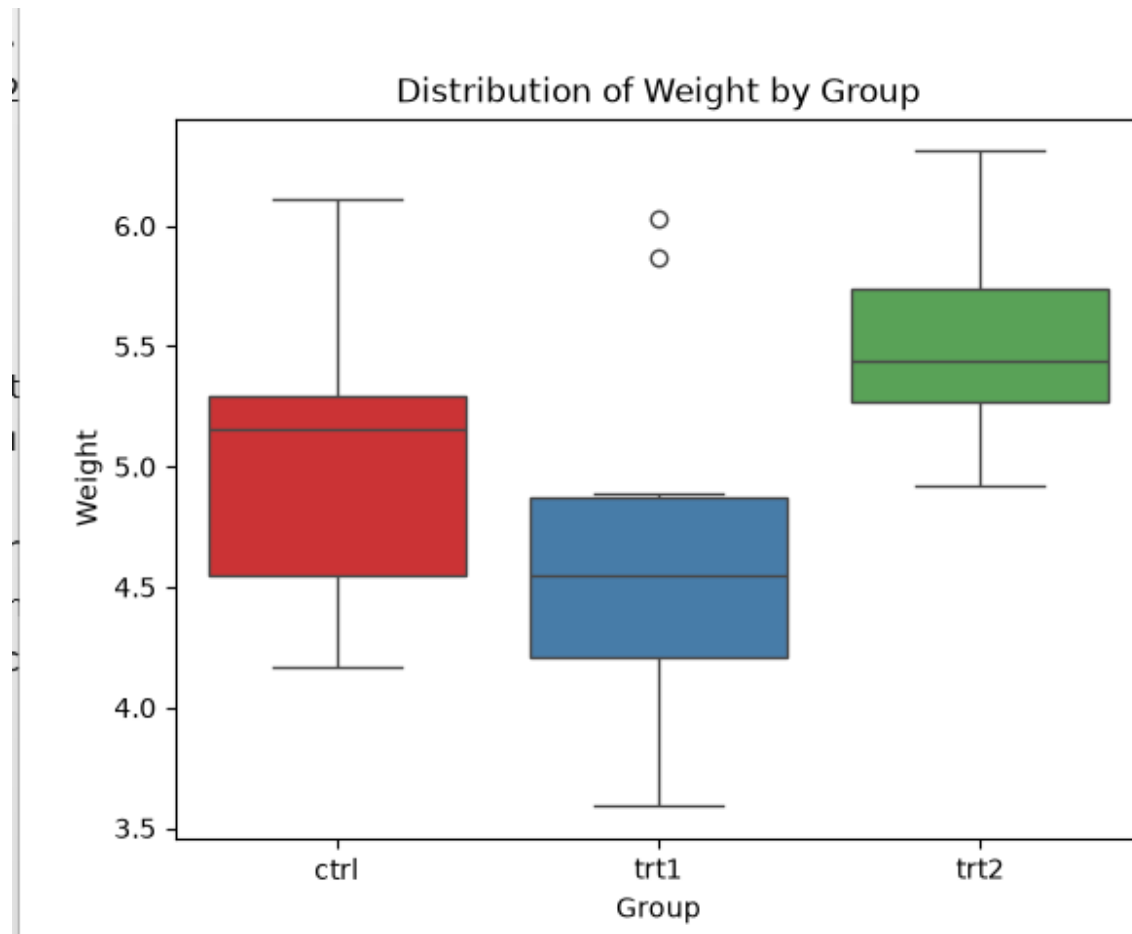
```
custom_bins = np.arange(3.3, 6.6, 0.3)
plt.hist(PlantGrowth["weight"], bins=custom_bins, color='red', alpha=0.7, edgecolor='black')
plt.xlabel("Weight")
plt.ylabel("Frequency")
plt.show()
```



b.

```
plt.boxplot(PlantGrowth["weight"])
plt.xlabel("Weight")
plt.show()
```

```
plant_df = pd.DataFrame(PlantGrowth)
sns.boxplot(x="group", y="weight", data=plant_df, palette="Set1")
plt.title("Distribution of Weight by Group")
plt.xlabel("Group")
plt.ylabel("Weight")
plt.show()
```



c. Based on the boxplot in 2b. I would approximate that 80% are below the minimum trt2 because there are two outliers and each group has 10 weights.

d.

```
trt1_weights = plant_df[plant_df["group"] == "trt1"]["weight"]
trt2_weights = plant_df[plant_df["group"] == "trt2"]["weight"]
min_trt2 = trt2_weights.min()
below_min = trt1_weights[trt1_weights < min_trt2]
percent_below_min = (len(below_min) / len(trt1_weights)) * 100
print(f"Percentage of trt1 weights below the minimum of trt2: {percent_below_min}%")
```

```
Percentage of trt1 weights below the minimum of trt2: 80.0%
```

```
filtered_df = PlantGrowth[PlantGrowth["weight"] > 5.5]
sns.barplot(data=filtered_df, x="group", y="weight", palette="deep")
plt.title("Average Weight per Group (Filtered > 5.5)")
plt.show()
```

e.

